平面图形的面积：

- 直角坐标系：

1. 曲线 $y=f(x)$，$y=g(x)$ $(f\le g)$ 以及直线 $x=a$，$x=b$ $(a<b)$ 所围成的面积：

$$A=\int_a^b (g(x)-f(x))\,dx$$

2. 曲线 $x=\varphi(y)$，$x=\psi(y)$ $(\varphi\le\psi)$ 以及直线 $y=c$，$y=d$ $(c<d)$ 所围成的面积：

$$A=\int_c^d (\psi(y)-\varphi(y))\,dy$$

3. 在1中，若去掉 $f\le g$ 的条件，则

$$A=\int_a^b |f(x)-g(x)|\,dx$$

- 例：求由曲线 $y=e^x$，$y=e^{-x}$ 及直线 $x=1$ 所围成图形的面积。

（板书）

- 例：求由曲线 $y^2=x$，$x-y-2=0$ 所围成的面积。

（板书）

- 求椭圆 $\frac{x^2}{a^2}+\frac{y^2}{b^2}=1$ 所围成的面积（$a>0$，$b>0$）。

提示：只用计算第一象限的面积 A_1。此时 $x=a\cos t$，$y=b\sin t$ $\left(0\le t\le\frac{\pi}{2}\right)$。$dA_1=y\,dx=b\sin t\cdot d(a\cos t)$

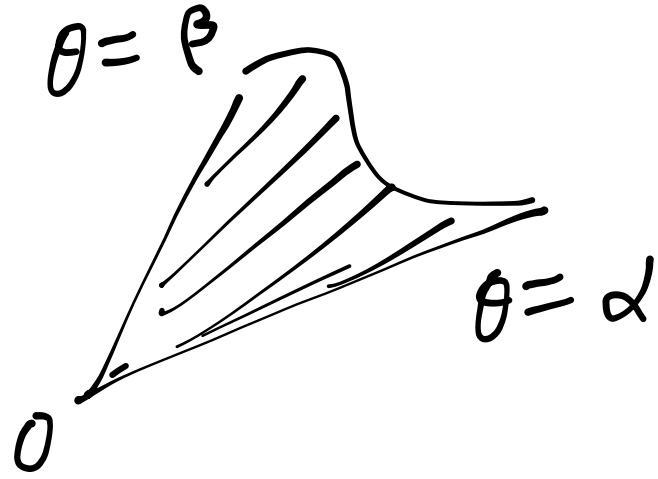
$=-ab\sin^2 t\,dt$

$A_1=\int_{\frac{\pi}{2}}^{0}-ab\sin^2 t\,dt=\frac{\pi}{4}ab$　$A=4A_1=\pi ab$

极坐标情形：

曲线 $\rho=\varphi(\theta)$. 在 $\alpha\le\theta\le\beta$ 上的曲边扇形面积.

$$A=\int_{\alpha}^{\beta}\frac{1}{2}\varphi^2(\theta)\,d\theta$$



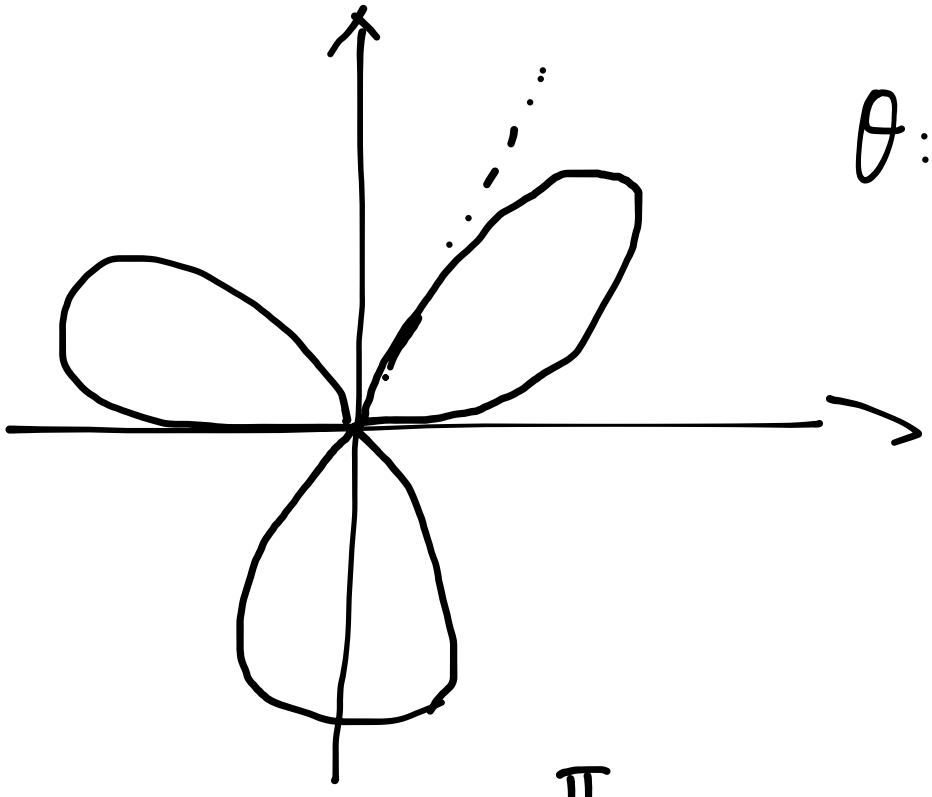
- 例. 求阿基米德螺线 $\rho=a\theta\ (a>0)$ 从 $0\to 2\pi$ 的一段弧与极轴所围的面积.

解: $A=\int_0^{2\pi}\frac{1}{2}\rho^2\,d\theta=\int_0^{2\pi}\frac{a^2}{2}\theta^2\,d\theta$

$=\frac{4}{3}\pi^3a^2$

□

例：计算三叶玫瑰线 $\rho = \sin 3\theta$ 的面积.



θ: $[0, \frac{\pi}{3}]$

$[\frac{2}{3}\pi, \pi]$

$[\frac{4}{3}\pi, \frac{5}{3}\pi]$

解：$S = 3\int_0^{\frac{\pi}{3}} \frac{1}{2}\rho^2 d\theta$

$= \frac{3}{2}\int_0^{\frac{\pi}{3}} \sin^2 3\theta \, d\theta$

$= \frac{3}{2}\int_0^{\frac{\pi}{3}} \frac{1-\cos 6\theta}{2} d\theta$

$= \frac{3}{2}\left(\frac{\theta}{2} - \frac{\sin 6\theta}{12}\right)\Big|_0^{\frac{\pi}{3}} = \frac{\pi}{4}$

∎

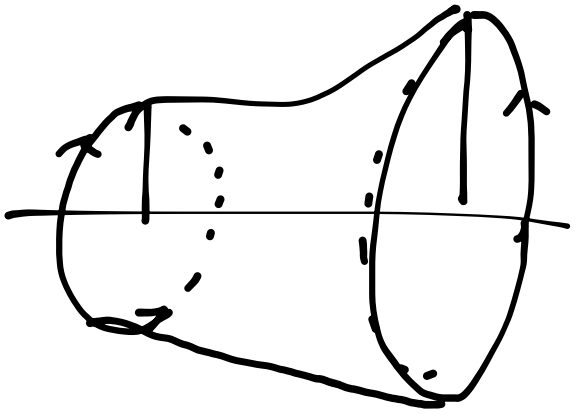
体积：

1. 旋转体的体积.

$y=f(x)$ 以及 $x=a$, $x=b$ 绕 x 轴旋转一周所成的体积：

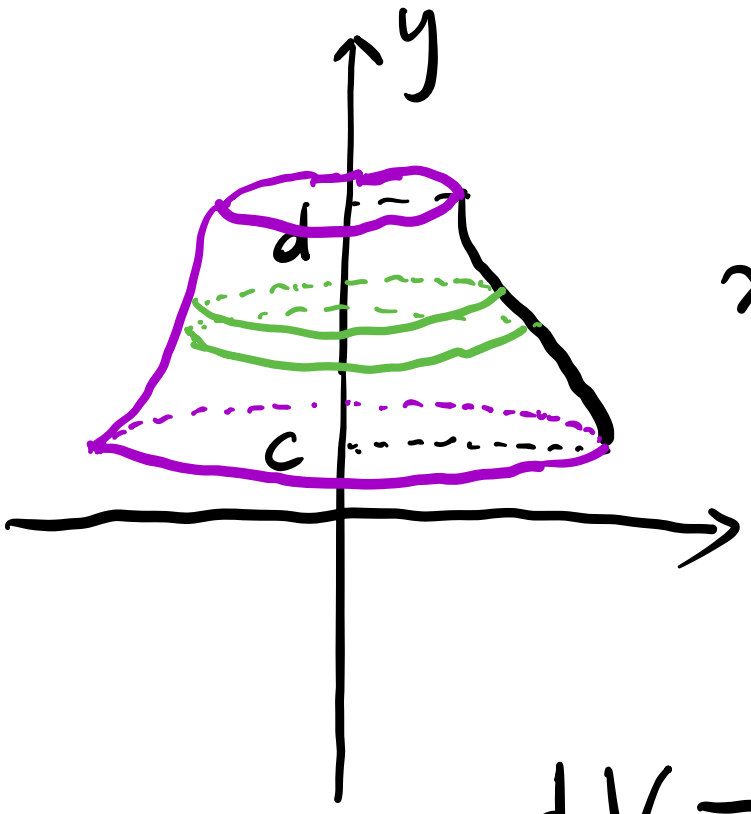
$$V=\int_a^b \pi f^2(x)\,dx$$

$$dV=\pi y^2 dx$$



$$V=\int_a^b \pi y^2 dx$$

$$y=f(x)$$



$$x = \varphi(y)$$

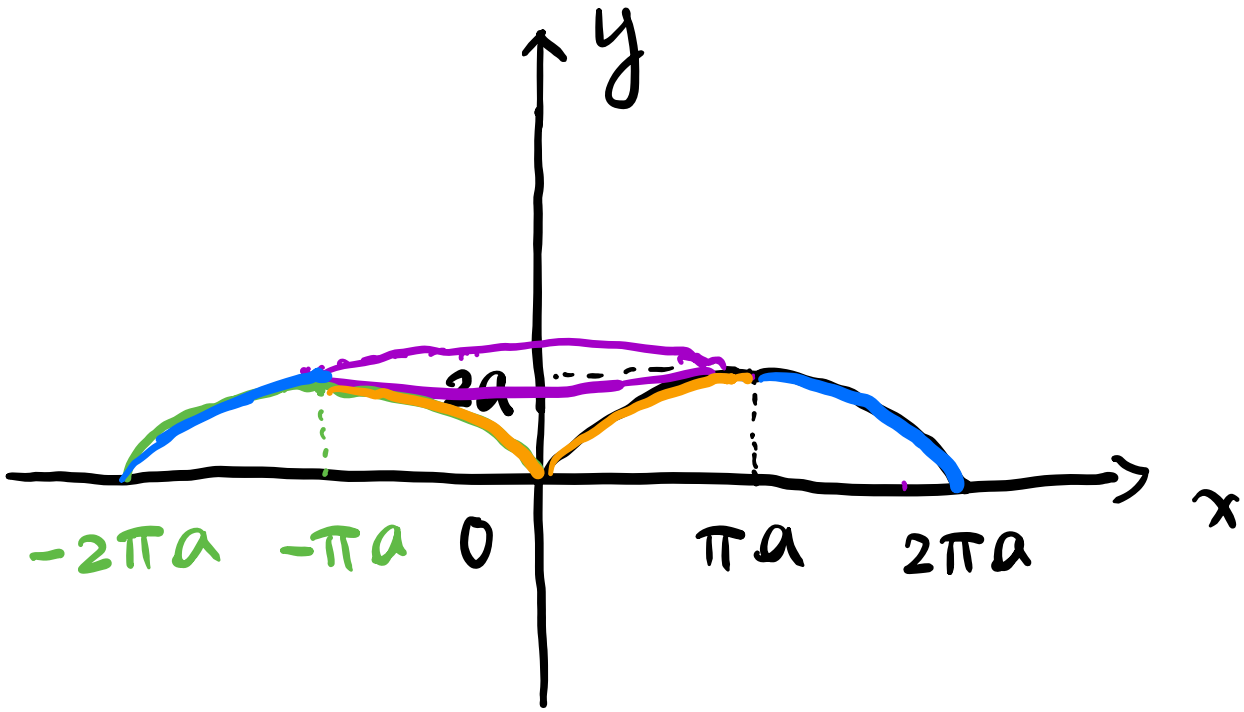
$$c \leq y \leq d$$

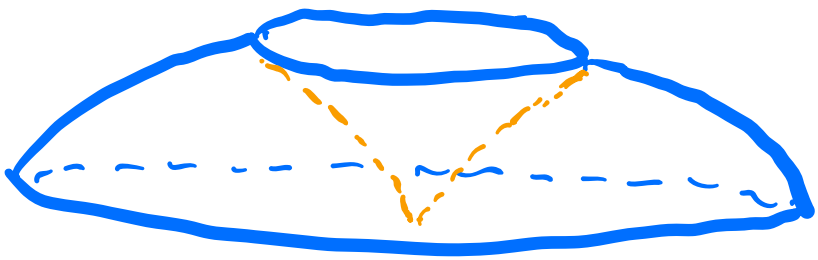
$$dV = \pi x^2 \, dy$$

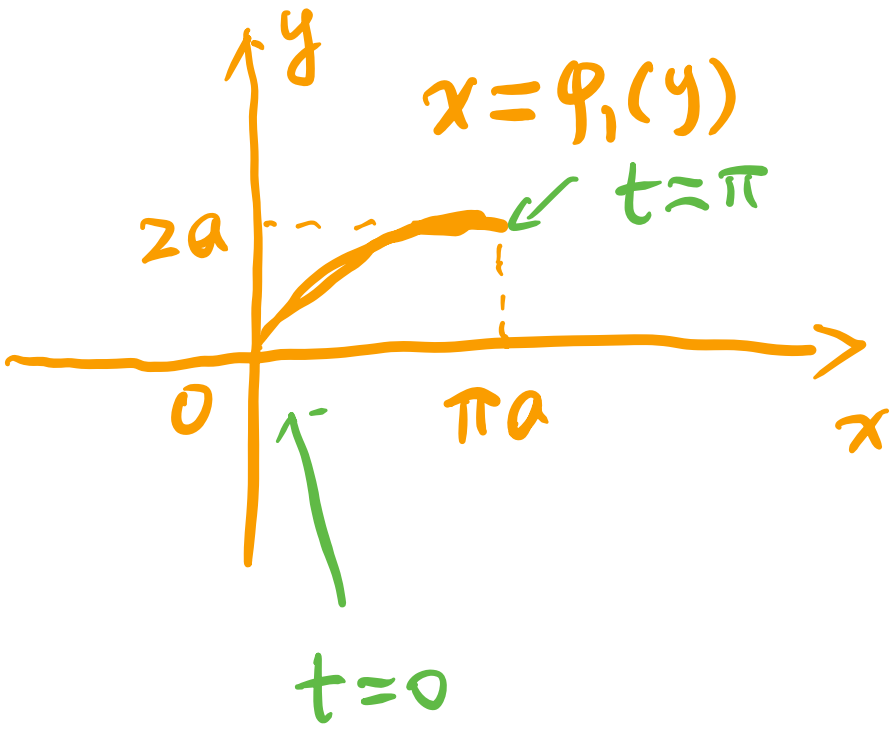
$$= \pi \varphi^2(y) \, dy$$

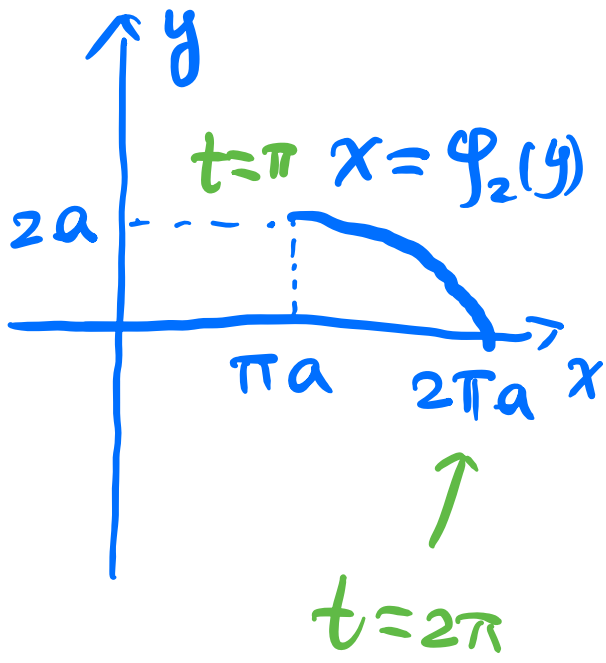
$$V = \int_c^d \pi \varphi^2(y) \, dy$$

$$V = \int_c^d \pi x^2 \, dy$$


y
2a
x
-2πa
-πa
0
πa
2πa




y
x = φ1(y)
t=π
2a
0
πa
x
t=0


y
t=π x = φ2(y)
2a
πa
2πa
x
t=2π

$$V = \int_0^{2a} \pi \varphi_2^2(y)\,dy - \int_0^{2a} \pi \varphi_1^2(y)\,dy$$

$t: 2\pi \to \pi$ $\qquad$ $t: 0 \to \pi$

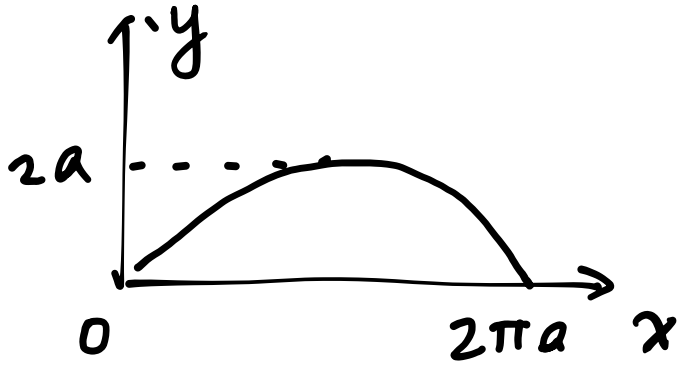
$$\begin{cases} x = a(t - \sin t) \\ y = a(1 - \cos t) \end{cases}$$

$t \in [0, \pi]$, $x = \varphi_1(y)$

$t \in [\pi, 2\pi]$, $x = \varphi_2(y)$

- 求摆线 $x=a(t-\sin t)$, $y=a(1-\cos t)$ 的一拱及 x 轴所围的图形分别绕 x 轴, y 轴旋转一周而成的体积.

解: (1) 绕 x 轴旋转一周围的体积



$$V_x=\int_0^{2\pi a}\pi y^2(x)\,dx$$

$$=\int_0^{2\pi}\pi a^2(1-\cos t)^2\,d\big(a(t-\sin t)\big)$$

$$=\pi a^3\int_0^{2\pi}(1-\cos t)^3\,dt$$

$$=\pi a^3\int_0^{2\pi}(1-3\cos t+3\cos^2 t-\cos^3 t)\,dt$$

积分为 0

$$=\pi a^3\int_0^{2\pi}\left(1+3\cdot\frac{1+\cos 2t}{2}\right)dt$$

$$=5\pi^2 a^3$$

绕 y 轴一圈的体积：

当 $0 \le x \le \pi a$ 时，摆线方程决定了一个函数 $x = x_1(y)$；当 $\pi a \le x \le 2\pi a$ 时，摆线方程决定了函数 $x = x_2(y)$.

$x = x_1(y)$ 对应于 $t: 0 \to \pi$; $y: 0 \to 2a$

$x = x_2(y)$ 对应于 $t: 2\pi \to \pi$; $y: 0 \to 2a$

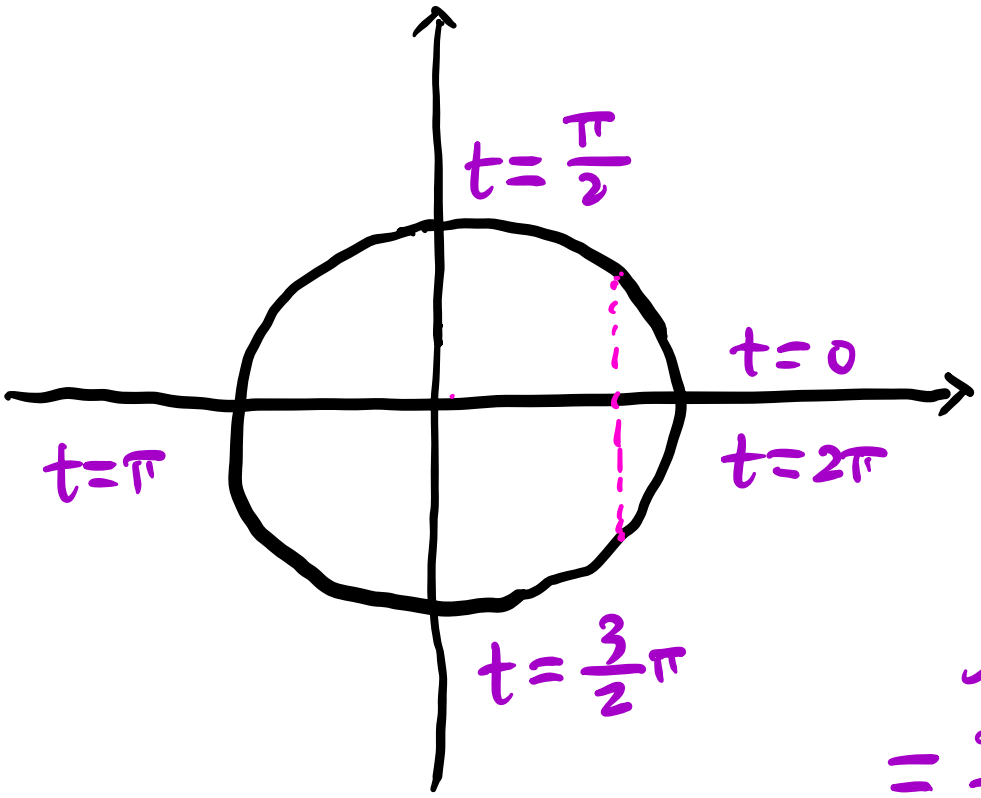
此时 $V_y = \int_0^{2a} \pi x_2^2(y)\,dy - \int_0^{2a} \pi x_1^2(y)\,dy$

$= \pi a^3 \left(\int_{2\pi}^{\pi} (t - \sin t)^2 \cdot \sin t\,dt - \int_0^{\pi} (t - \sin t)^2 \sin t\,dt \right) = -\pi a^3 \int_0^{2\pi} (t - \sin t)^2 \sin t\,dt$

$= 6\pi^3 a^3$

($\int_0^{2\pi} \sin^3 t\,dt = 0$, $\int_0^{2\pi} t \sin^2 t\,dt = \pi \int_0^{2\pi} \sin^2 t\,dt$

$\int_0^{2\pi} t^2 \sin t\,dt$ 分部积分)

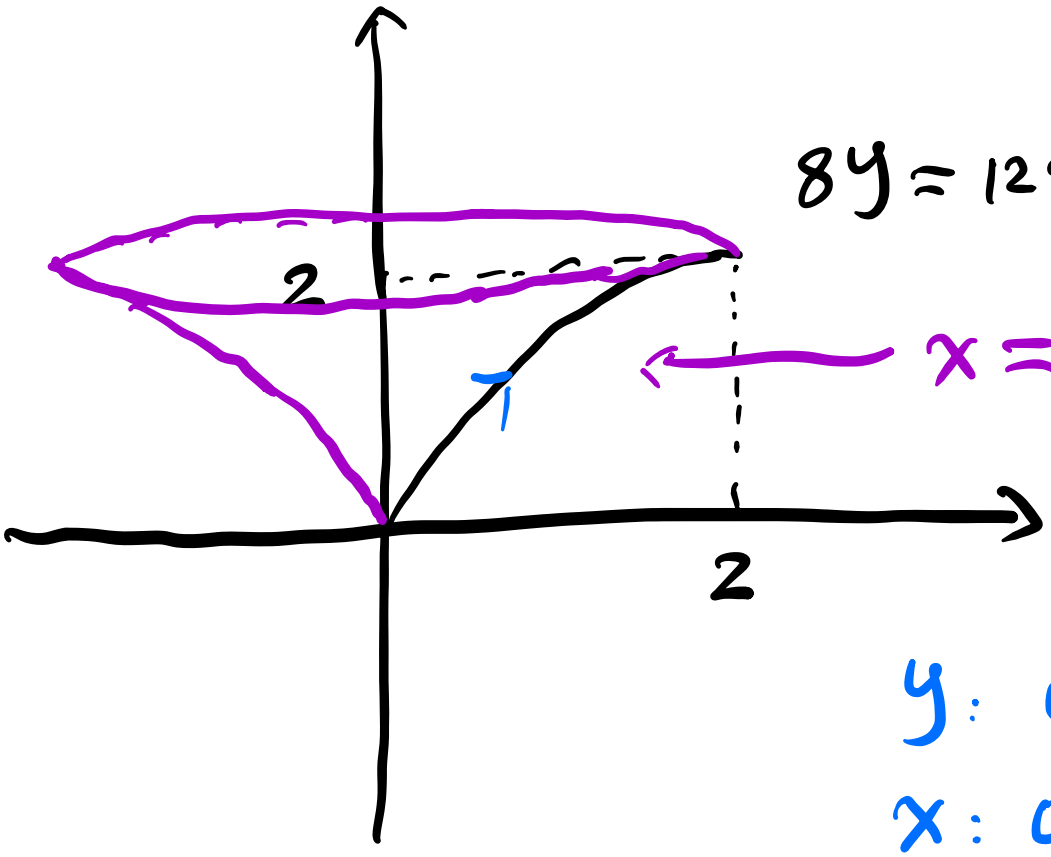


$$\int_0^{\pi} x f(\sin x)\,dx = \frac{\pi}{2}\int_0^{\pi} f(\sin x)\,dx$$

$$\int_0^{2\pi} \sin^3 t\,dt = 0$$

$$\int_0^{2\pi} t\sin^2 t\,dt = \pi\int_0^{2\pi} \sin^2 t\,dt$$

书上例5.7:



$8y = 12x - x^3$

$x = \varphi(y)$

$y: 0 \to 2$

$x: 0 \to 2$

$$V = \int_0^2 \pi x^2 \, dy$$

变量代换

$$= \int_0^2 \pi x^2 \, d\frac{12x - x^3}{8}$$

- 平面图形 $0 \le a \le x \le b$,

$0 \le y \le f(x)$ 绕 y 轴旋转成的

旋转体的体积为

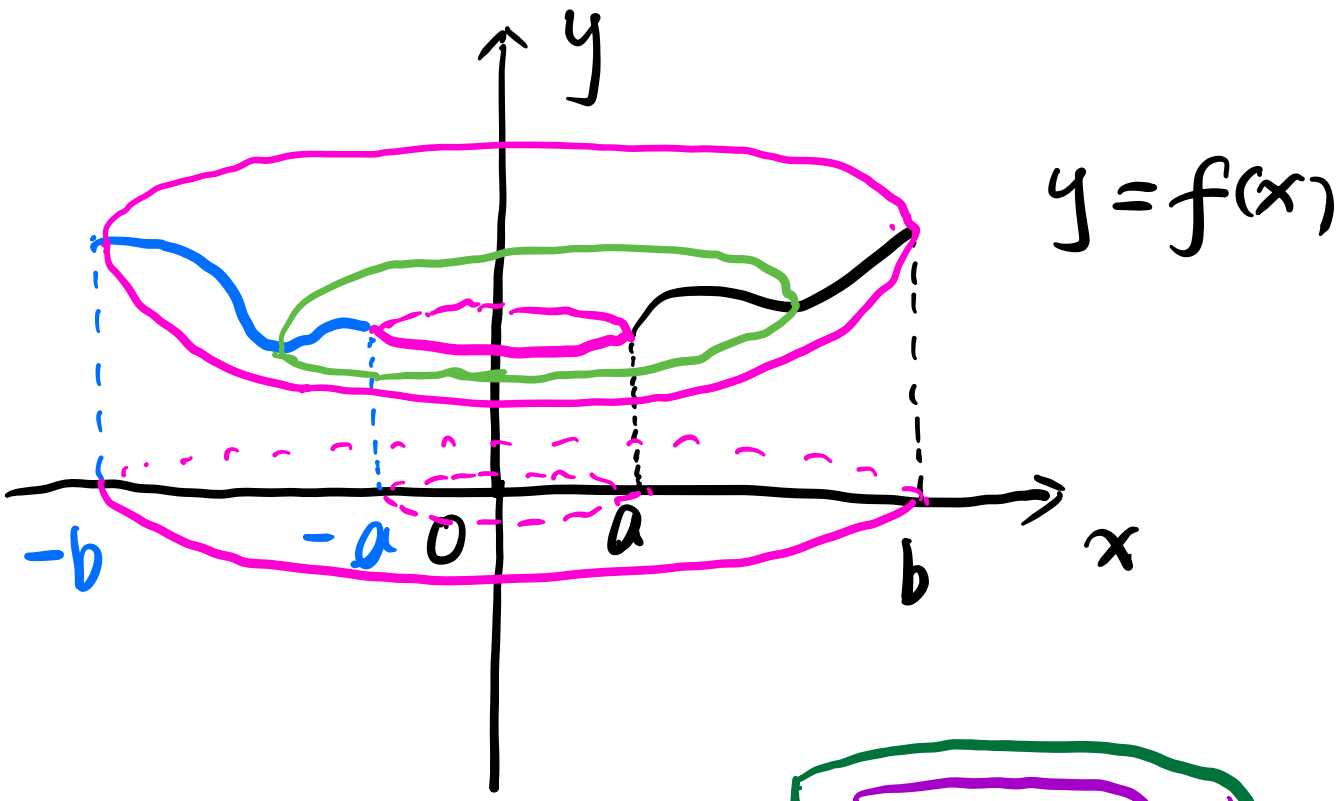
$$V = 2\pi \int_a^b x f(x)\,dx$$

(柱壳法)

书上 P261 页柱壳法.

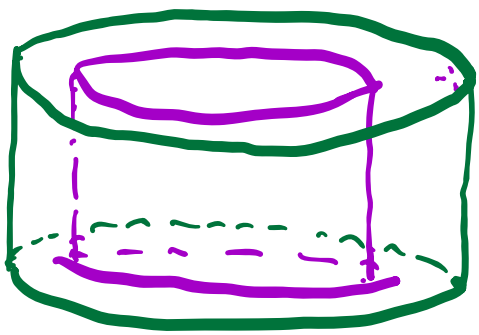
$2\pi x\,dx$ 底面同心圆圈的面积元

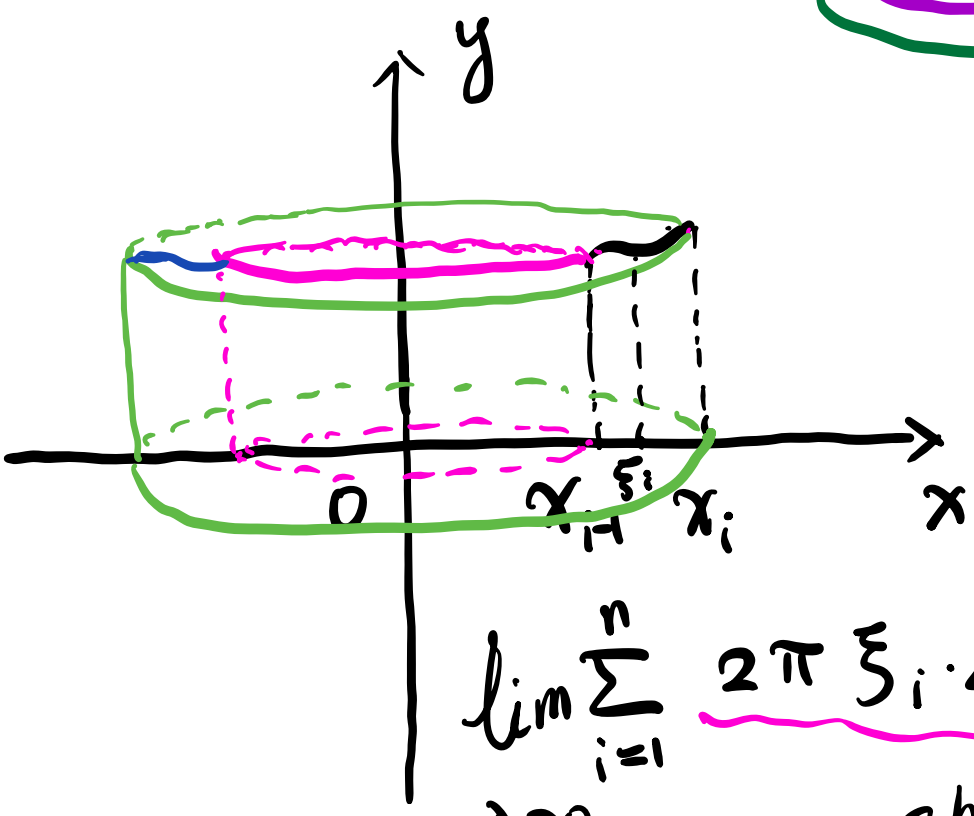
$f(x)$ 高度



$y=f(x)$

柱壳法 →



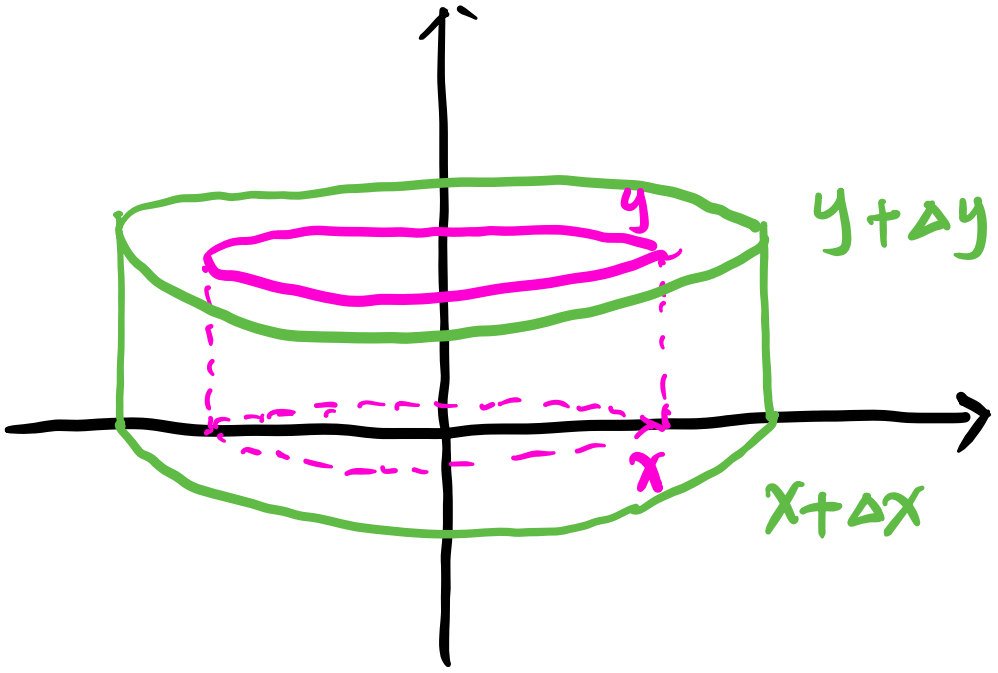




$a=x_0<x_1<\cdots<x_n$

$$\lim_{\lambda\to 0}\sum_{i=1}^{n} 2\pi \xi_i \cdot \Delta x\, f(\xi_i)$$

$$=\int_a^b 2\pi x f(x)\,dx$$



$$\Delta V \approx y\left[\pi(x+\Delta x)^2 - \pi x^2\right]$$

$$= y\left(2\pi x\cdot \Delta x + \pi \Delta^2 x\right)$$

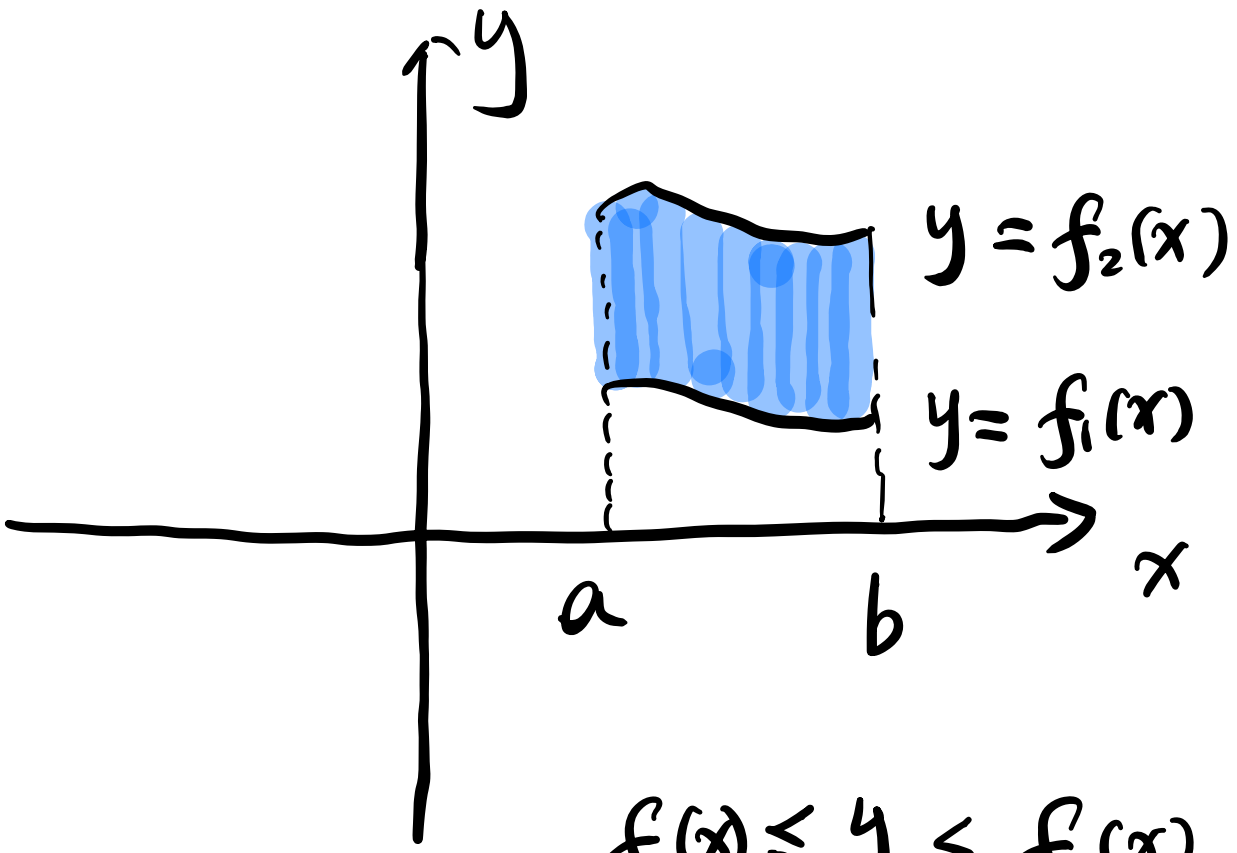
$$= 2\pi x\, y\cdot \Delta x + \pi y\cdot \Delta^2 x$$

dV $\qquad$ $o(\Delta x)$

$$\boxed{dV = 2\pi x\, y\, dx}$$

$f(x)\geq 0$
$b\geq a\geq 0$

$$= 2\pi x f(x)\, dx$$



$f_1(x) \le y \le f_2(x)$

区域 $a \le x \le b$ $(a \ge 0)$

绕y轴转一圈围成的体积

$$V = \int_a^b 2\pi x \left(f_2(x) - f_1(x)\right) dx$$

$2\pi x\,dx$ 柱壳底面积微元

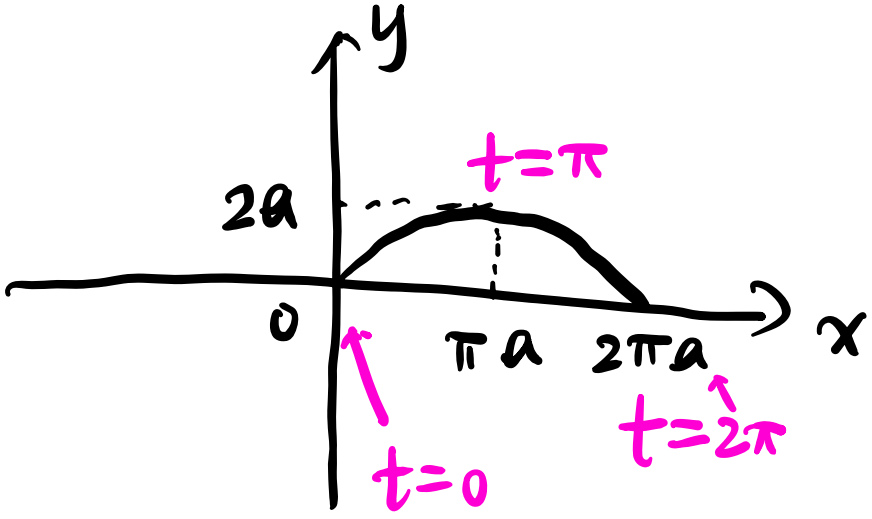
$f_2(x) - f_1(x)$ 柱壳高度

前面的例子：

求摆线 $\begin{cases} x = a(t - \sin t) \\ y = a(1 - \cos t) \end{cases}$

$t \in [0, 2\pi]$

和 x 轴所围成的部分绕 y 轴旋转一圈所围成的几何体的体积.



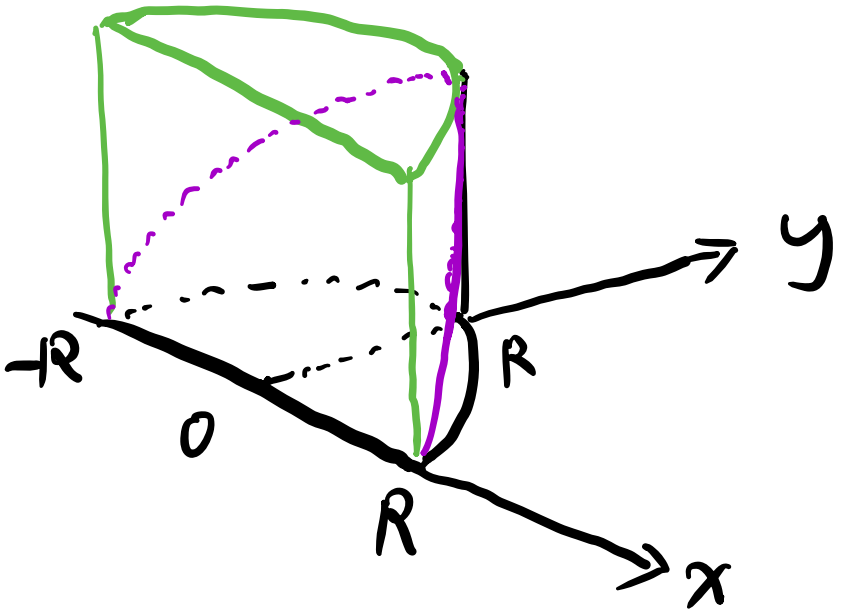
柱壳法：$V = \int_0^{2\pi a} 2\pi x \cdot y \, dx$

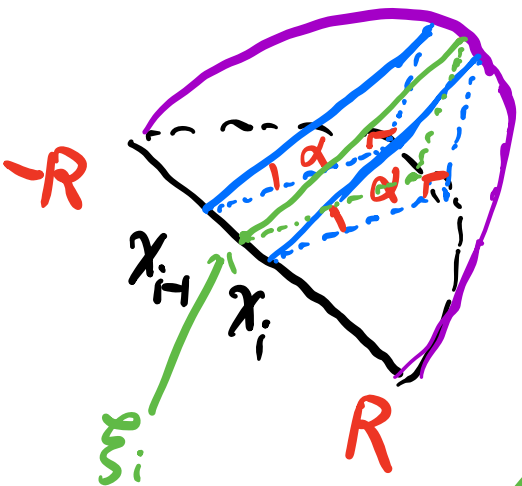
$$= \int_0^{2\pi} 2\pi a(t-\sin t)\cdot a(1-\cos t)\, d\big(a(t-\sin t)\big)$$

$$= \int_0^{2\pi} 2\pi a^3 (t-\sin t)\cdot(1-\cos t)^2\, dt$$

（计算过程略）

书上例5.8

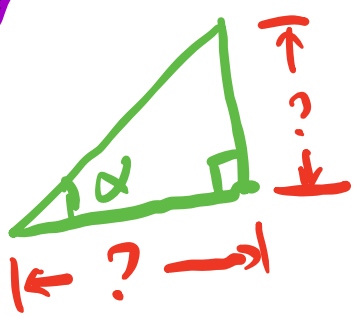




任意分割

$-R=x_0<x_1<\cdots<x_n=R$

$\forall\ \xi_i\in[x_{i-1},x_i]$



$\sqrt{R^2-\xi_i^2}\cdot\tan\alpha$



$\sqrt{R^2-\xi_i^2}$

$S_\Delta=\frac{1}{2}\left(R^2-\xi_i^2\right)\cdot\tan\alpha$

$$V=\lim_{\lambda\to 0}\sum_{i=1}^{n}\frac{1}{2}(R^2-\xi_i^2)\cdot\tan\alpha\cdot\Delta x_i$$

$$=\int_{-R}^{R}\frac{1}{2}(R^2-x^2)\cdot\tan\alpha\cdot dx$$

(略)

曲线弧长：

1. $y=f(x)\quad (a\le x\le b)$

$$\begin{cases} x=x \\ y=f(x) \end{cases}$$

x为参数

$$S=\int_a^b \sqrt{1+(y')^2}\,dx \qquad ds$$

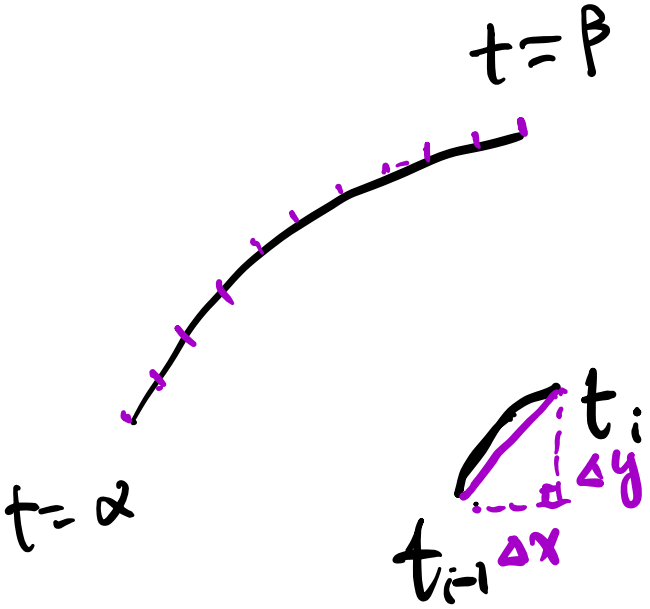
2. $x=\varphi(t),\ y=\psi(t)\quad (\alpha\le t\le\beta)$

$$S=\int_\alpha^\beta \sqrt{(\varphi'(t))^2+(\psi'(t))^2}\,dt \qquad ds$$

3. $\rho=\rho(\theta)\quad (\alpha\le\theta\le\beta)$

$$S=\int_\alpha^\beta \sqrt{\rho^2(\theta)+(\rho'(\theta))^2}\,d\theta \qquad ds$$

$$\begin{cases} x=\rho(\theta)\cos\theta \\ y=\rho(\theta)\cdot\sin\theta \end{cases}$$



$x = x(t)$

$y = y(t)$

$\alpha = t_0 < t_1 < \cdots < t_n = \beta$

$\xi_i \in [t_{i-1}, t_i]$

$$\Delta s \approx \sqrt{(\Delta x)^2 + (\Delta y)^2}$$

$$\approx \sqrt{(x')^2 + (y')^2} \cdot \Delta t$$

在 $[t_{i-1}, t_i]$ 中的某点取值

$$\int_\alpha^\beta \sqrt{(x')^2 + (y')^2}\, dt$$

曲线长度

$$ds = \sqrt{(x')^2 + (y')^2}\, dt$$

$$\Delta s \approx \sqrt{(x')^2 + (y')^2} \cdot \Delta t$$

• 例：求 $y=\frac{e^x+e^{-x}}{2}$ 从 $x=0$ 到 $x=a$ $(a>0)$ 那一段弧的弧长.

ds

解：$s=\int_0^a \sqrt{1+(y')^2}\,dx$

$=\int_0^a \sqrt{1+\operatorname{sh}^2 x}\,dx=\int_0^a \operatorname{ch} x\,dx$

$=\operatorname{sh} x\Big|_0^a=\operatorname{sh} a=\frac{e^a-e^{-a}}{2}$

• 求摆线 $x=a(t-\sin t)$，$y=a(1-\cos t)$ 的一拱的长度 $(0\le t\le 2\pi)$ $(a>0)$

解：$s=\int_0^{2\pi} \sqrt{(x'(t))^2+(y'(t))^2}\,dt$

$=\int_0^{2\pi} a\sqrt{(1-\cos t)^2+\sin^2 t}\,dt$

$=\int_0^{2\pi} a\sqrt{2-2\cos t}\,dt$

$$= a\int_0^{2\pi} 2\sin\frac{t}{2}\,dt = -4a\cos\frac{t}{2}\Big|_0^{2\pi} = 8a$$

- 例：求心脏线 $\rho = a(1+\cos\theta)$ 一周全长.

$(0 \le \theta \le 2\pi)$ $(a>0)$

解：$$S = \int_0^{2\pi} \sqrt{\rho^2(\theta) + (\rho'(\theta))^2}\,d\theta$$

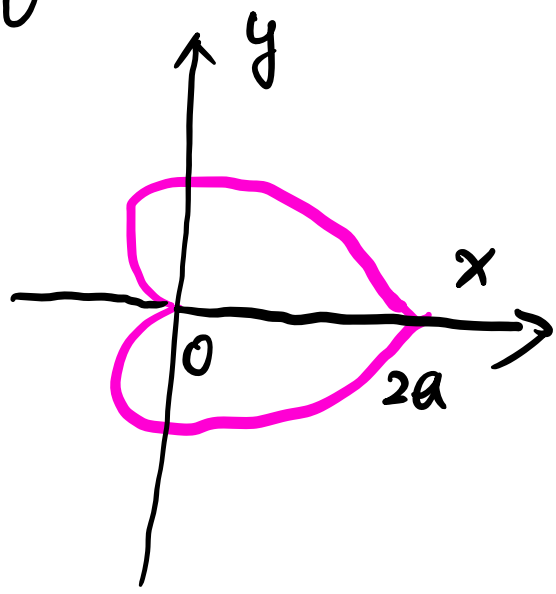
$$= a\int_0^{2\pi} \sqrt{(1+\cos\theta)^2 + (-\sin\theta)^2}\,d\theta$$

$$= a\int_0^{2\pi} \sqrt{2+2\cos\theta}\,d\theta$$

$$= a\int_0^{2\pi} 2\left|\cos\frac{\theta}{2}\right| d\theta$$

$$= 4a\int_0^{\pi} \cos\frac{\theta}{2}\,d\theta$$

$$= 8a\sin\frac{\theta}{2}\Big|_0^{\pi} = 8a$$



作业：

习题5-5A类：

1. (2).

2. (1)

3. (1)

4.

5.

10.

11.

习题5-5B类

6.

定积分在物理中的应用

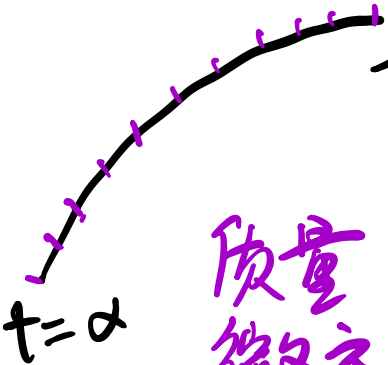
曲线质量：m



$$\begin{cases} x = x(t) \\ y = y(t) \end{cases}$$

$$t \in [\alpha, \beta]$$

线密度：$\rho = \rho(t)$



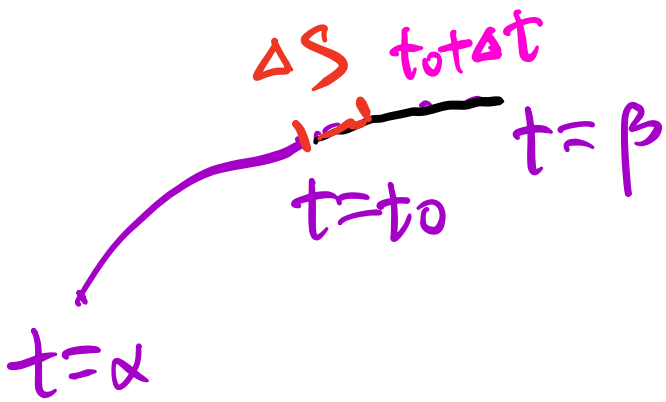
$$ds = \sqrt{(x'(t))^2 + (y'(t))^2}\,dt$$

质量微元

$$dm = \rho(t)\,ds$$

$$= \sqrt{(x'(t))^2 + (y'(t))^2}\,dt$$

设 $m(t_0)$ 为曲线从 $t=\alpha$ 到 $t=t_0$ 之间的曲线的质量



$$\rho(t_0)=\lim_{\Delta t\to 0}\frac{m(t_0+\Delta t)-m(t)}{\Delta S}$$

$$=\lim_{\Delta t\to 0}\frac{\Delta m}{\Delta S}$$

$$m=\int_{\alpha}^{\beta}\rho(t)\cdot\sqrt{(x'(t))^2+(y'(t))^2}\,dt$$

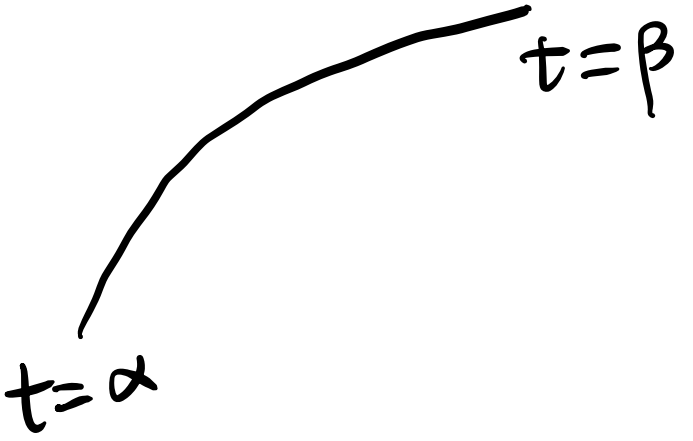
$y=f(x)\quad a\le x\le b$

$\rho=\rho(x)$

$\begin{cases}x=x\\ y=f(x)\end{cases}$ x为参数

$$m=\int_{a}^{b}\rho(x)\cdot\sqrt{1+(f'(x))^2}\,dx$$

曲线质心坐标



$$\begin{cases} x = x(t) \\ y = y(t) \end{cases}$$

$t \in [\alpha, \beta)$

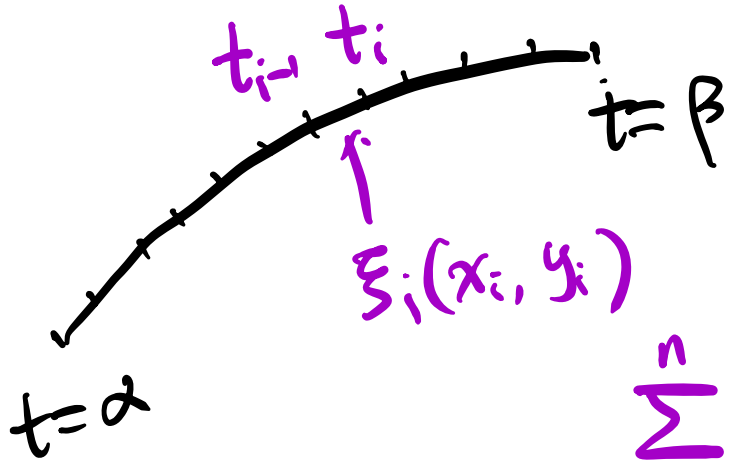
$\rho = \rho(t)$

离散的质心公式：n个点

$P_i(x_i, y_i)$，质量 m_i　　$1 \le i \le n$

质心

$$\begin{cases} \bar{x} = \dfrac{\sum_{i=1}^{n} m_i x_i}{\sum_{i=1}^{n} m_i} \\ \bar{y} = \dfrac{\sum_{i=1}^{n} m_i y_i}{\sum m_i} \end{cases}$$



$\alpha = t_0 < t_1 < \cdots < t_n = \beta$

$\xi_i(x_i, y_i)$

$$\bar{x} = \frac{\sum_{i=1}^{n} \Delta m_i \cdot x_i}{\sum_{i=1}^{n} \Delta m_i}$$

$$\approx \frac{\sum_{i=1}^{n} \rho(\xi_i) \cdot \Delta s_i \cdot x_i}{\sum_{i=1}^{n} \rho(\xi_i) \cdot \Delta s_i}$$

令 $\lambda \to 0$

$$\bar{x} = \frac{\int_{\alpha}^{\beta} \rho(t) \cdot x(t)\, ds}{\int_{\alpha}^{\beta} \rho(t)\, ds}$$

$$= \frac{\int_\alpha^\beta \rho(t)\, x(t) \sqrt{(x'(t))^2+(y'(t))^2}\, dt}{\int_\alpha^\beta \rho(t) \cdot \sqrt{(x'(t))^2+(y'(t))^2}\, dt}$$

$$\bar{y} = \frac{\int_\alpha^\beta \rho(t)\, y(t) \sqrt{(x'(t))^2+(y'(t))^2}\, dt}{\int_\alpha^\beta \rho(t) \cdot \sqrt{(x'(t))^2+(y'(t))^2}\, dt}$$

若 $\rho(t)$ 为常数，则

$(\bar{x}, \bar{y})$ 为重心坐标.

此时

$$\tilde{x}=\frac{\int_{\alpha}^{\beta} x(t)\sqrt{(x'(t))^2+(y'(t))^2}\,dt}{\int_{\alpha}^{\beta}\sqrt{(x'(t))^2+(y'(t))^2}\,dt}$$

$$\tilde{y}=\frac{\int_{\alpha}^{\beta} y(t)\sqrt{(x'(t))^2+(y'(t))^2}\,dt}{\int_{\alpha}^{\beta}\sqrt{(x'(t))^2+(y'(t))^2}\,dt}$$